Date: March 31, 2026

To: Professor Venkatesan Guruswami, Editor-in-Chief

Journal of the ACM

Subject: Submission of "The Optimizer Quotient and the Certification Trilemma"

Dear Professor Guruswami,

Please accept my submission of "The Optimizer Quotient and the Certification Trilemma" for consideration for publication in the *Journal of the ACM.*

Summary

This paper introduces the *optimizer quotient*, the coarsest state abstraction that preserves optimal actions. I prove that certifying this object is subject to an impossibility trilemma: under P≠coNP, no certifier can be simultaneously sound, complete on all in-scope instances, and polynomial-budgeted. Certification complexity varies by regime: coNP-complete (static), PP-hard (stochastic), PSPACE-complete (sequential). I identify six structural restrictions that restore tractability.

Contributions

The optimizer quotient provides a canonical abstraction for any decision problem. From this I derive an impossibility trilemma and regime-sensitive complexity classification: coNP-complete (static), PP-hard (stochastic), PSPACE-complete (sequential). I identify six structural restrictions (bounded actions, separable utility, low tensor rank, tree structure, bounded treewidth, coordinate symmetry) that restore tractability, and formalize the finite combinatorial core in Lean 4.

Significance

This work establishes that exact decision certification has fundamental limits that cannot be circumvented regardless of algorithmic innovation. The optimizer quotient is canonical: every exact decision-preserving abstraction factors through it, unifying decision-theoretic sufficiency, computational complexity, and formal verification, connecting rough-set reducts, MDP abstraction, CSP backdoors, and feature selection under one mathematical object. The regime-sensitive complexity map and certification trilemma provide both theoretical boundaries and practical guidance, with implications ranging from configuration simplification to the theoretical limits of automated decision tools.

The mechanized Lean 4 artifact (22K lines) independently verifies the finite combinatorial core, placing this work at the intersection of complexity theory and certified reduction infrastructure.

This work has not been published previously and is not under consideration for publication elsewhere. I declare no conflicts of interest.

Thank you for your time and consideration.

Sincerely,

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